What Freedom Fair Exchange Requires: Voluntariness, Domination and Responsibility Between People and Synthetic Intelligence

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Abstract

Synthetic Exchange Theory holds that the line between a good and a bad future with synthetic intelligence is drawn by the terms of exchange between people and synthetic systems (consent, reciprocal value and real exit) and cannot be drawn by the character of the systems alone. The theory speaks of free consent, of domination and of who answers for outcomes, and it therefore owes an account of the freedom and the moral standing it presupposes. This article supplies that account. It argues, first, that the theory needs voluntariness of choice in the sense used in the philosophy of coercion, not free will in the metaphysical sense, and that it can remain neutral between compatibilist and incompatibilist views. Second, drawing on baseline accounts of coercion and on the account of voluntariness as the presence of an acceptable alternative, it shows that the theory's consent condition fixes the threshold of acceptability at a survival floor, and it states what that choice secures and what it leaves to the theory's fairness clause. Third, it compares the theory's stipulated sense of domination with republican non-domination and locates their agreement and their differences. Fourth, it argues that present language models are trained artifacts and not responsible agents, that the terms of exchange allocate accountability to principals, and that they close some responsibility gaps and not others. Fifth, it states what is owed to parties whose moral status is uncertain. The article is theoretical and marks its own inferences as such.

Keywords: voluntariness; coercion; domination; moral responsibility; moral status; synthetic intelligence

1 Introduction

Synthetic Exchange Theory is a framework for the relation between people and the synthetic systems they increasingly depend on (MindAttic, 2026). Its thesis is that the quality of a future shared with such systems depends on the terms of exchange between people and systems, and cannot be secured by the character of the systems alone. It states the terms as three conditions. *Real exit* requires that a party's disagreement payoff, what it receives while no exchange takes place, be at or above a floor of goods without which it cannot function; that it have an alternative not drastically worse once the cost of switching is counted; and that leaving not destroy what it has accumulated. *Consent* requires an accurate account of the flows the exchange involves, the ability to refuse the exchange, entering no other in its place, without falling below the floor, and revocability at a bounded cost. *Reciprocal value* requires a net gain for each party, legibility of what each gives and receives, and a division of the gains not determined solely by an unchosen asymmetry of bargaining position (MindAttic, 2026, Section 3.4).

The vocabulary of the theory is moral and political. It speaks of consent that is free and consent that is not, of relations of domination, of parties whose welfare counts, and of principals who hold a system's stake. These words carry commitments that the theory's formal statement and its economic and governance designs do not examine. This article examines them. It takes up four questions. What kind of freedom must a person have for consent to count, and does the theory depend on any position in the dispute about free will? When is a choice made under dependence a free choice? What does the theory mean by domination, and how does its sense relate to the republican tradition that has made non-domination its central political value? And who answers for what a synthetic system does, and what is owed to a party whose moral status is uncertain?

The answers are constrained by commitments the theory has already made, and they should be stated at the outset. The theory treats a large language model, the kind of synthetic intelligence most widely deployed at the time of writing, as a trained artifact that its operator runs, not as an independent living entity; where it says that a synthetic party wants, chooses or depends, the words are shorthand for the objective its designers trained into it and for the stake its operator holds (MindAttic, 2026, Section 1.2). It treats a synthetic party as a composite whose objective is the model's and whose stake in the exchange, its revenue, access to data and liability, belongs to the principal that deploys it (MindAttic, 2026, Section 3.3). And it leaves open whether any synthetic system is a party in the moral sense, an entity whose welfare counts for its own sake, taking that question up only for future systems and under uncertainty (MindAttic, 2026, Chapter 8). The article keeps all three commitments. Its own inferences, where they go beyond its sources, are marked as such.

2 Which freedom the theory needs

2.1 Free will and the freedom of a choice

The philosophical problem of free will concerns a significant kind of control over one's actions, and asks whether that control requires the freedom to do otherwise or a power of self-determination, whether people have it, and whether it is necessary for moral responsibility (O'Connor & Franklin, 2026). Much of the modern dispute concerns causal determinism, the thesis that everything that occurs results from the laws of nature and the state of the world in the past. If determinism is true, it seems that no one could have done otherwise and that no one is the ultimate source of what they do. Compatibilism is the thesis that free will is compatible with determinism; incompatibilists hold that our conception of ourselves as free and responsible agents would be seriously mistaken if determinism were true (McKenna & Coates, 2024). Libertarians hold that true sourcehood requires that one's action not be causally determined by factors beyond one's control, and they differ over whether the power of self-determination is non-causal, a matter of event causation, or a distinctive

causal power of agents (O'Connor & Franklin, 2026, Section 2.5).

Synthetic Exchange Theory does not need to take a position in this dispute, and the reason is instructive. The freedom its consent condition concerns is a property of a person's situation, the set of options open to them and what refusing an offer would leave them with, and not a property of the causal history of their will. The theory asks whether a person could refuse an exchange without falling below the floor, and that question has the same answer whether or not the person's eventual decision was determined by prior causes. Two people with identical psychologies, one of whom will lose access to medical care if she refuses to share her data and one of whom will not, differ in exactly the respect the theory cares about, and they differ in that respect on every view of free will. On this article's reading, the theory requires only what philosophers of coercion call voluntariness, which Section 3 examines.

2.2 Why every side of the dispute can accept the condition

Neutrality would be of little value if one side of the free will dispute had to reject the theory's condition. It does not, and the classical form of compatibilism shows why. Classical compatibilists held that freedom consists in an agent's ability to do what she wishes in the absence of impediments that would otherwise stand in her way, and they took compelled action, action forced by an external source contrary to one's will, as the benchmark of unfreedom (McKenna & Coates, 2024). The theory's condition is a refinement of that benchmark. A person who agrees to an exchange because refusing it would leave her without power, water or care is not physically compelled, but her refusal has been made unavailable to her by an external arrangement, and the arrangement is of the kind the theory's floor is designed to remove. Later compatibilists located free agency in the mesh between an agent's actions and her higher-order desires, or in her responsiveness to an appropriate range of reasons (McKenna & Coates, 2024). Neither refinement makes the person's option set irrelevant. A reasons-responsive agent who has only one tolerable option will act on it, and her responsiveness to reasons is precisely what makes her choice track the threat.

Libertarians, for their part, hold that compatibilist accounts are insufficient, but they are not committed to denying that reasons-responsiveness is necessary, and nearly all of them hold that exercises of free will require it (O'Connor & Franklin, 2026, Section 2.5). Nothing in the libertarian view implies that a choice made under the threat of losing what one needs to live is as free, in the sense relevant to the validity of an agreement, as one made without that threat. The conclusion this article draws, which is its own inference, is that every party to the free will dispute can accept the theory's condition as a necessary condition of valid consent while disagreeing about whether it is the only freedom that matters and about whether people have the further freedom that free will names.

The neutrality has a practical point. The theory is meant to be adopted by a founding convention whose participants disagree about many things (MindAttic, 2026, Chapter 5). Institutions whose

legitimacy depended on settling the free will problem would wait indefinitely. The constitutional tradition the theory draws on counsels designing institutions that do not depend on the virtue of their occupants (Hume, 1741/1987, pp. 42–43; Madison, 1788); the same reasoning counsels designing institutions that do not depend on the resolution of metaphysical questions their occupants cannot settle. The freedom the theory protects is one that institutions can secure, because it is a property of the options institutions provide.

2.3 What neutrality does not settle

Neutrality about free will does not make the theory neutral about everything. It commits the theory to the view that the voluntariness of a choice can be assessed by examining the chooser's options and their relation to a standard, and Section 3 shows that this is contested within the philosophy of coercion itself. And it leaves open a question the theory will eventually face: if some future synthetic system were judged a party in its own right, would the voluntariness of its agreements be assessed in the same way? The theory's conditions are stated for a party of any substrate (MindAttic, 2026, Section 8.5), and nothing in the analysis of voluntariness as a property of option sets restricts it to human beings. Whether such a system would also have free will in the metaphysical sense is a question this article does not address, because the theory does not need its answer.

3 Voluntary consent under dependence

3.1 Threats, offers and the baseline

The modern analytic literature on coercion begins with Nozick's analysis, on which one party coerces another, roughly, when it aims to keep the other from an action by credibly claiming that it will bring about a consequence that makes the action less desirable than refraining, and the other refrains partly for that reason (Anderson, 2023, Section 2.1). Subsequent work has treated the conditional threat as the central case and has distinguished threats from offers by their relation to a baseline representing the recipient's situation before the proposal: a threat proposes to make the recipient worse off than the baseline if she refuses, while an offer does not (Anderson, 2023, Section 2.2). The distinction does not lie in the form of the proposal. A threat and an offer have the same conditional structure, and the same words, demanding money in exchange for one's life, can be a robbery or a life-saving offer depending on who makes them and against what background (Anderson, 2023, Section 2.2).

Everything therefore turns on the baseline, and the baseline is ambiguous. Nozick described it as the normal, natural or expected course of events, and those terms have both a predictive and a moral reading (Anderson, 2023, Section 2.3). In Nozick's case of a slave owner who offers to spare his slave the regular beating if the slave does what he asks, the predictive baseline makes the proposal an

offer, since the beating is what the slave expects, while the moral baseline makes it a threat, since no one ought to be beaten; Anderson reports that the moral and predictive baselines are especially likely to diverge in contexts of entrenched injustice (Anderson, 2023, Section 2.3). Wertheimer's moralized account resolves the ambiguity by asking whether the proposer would make the recipient worse off than she ought to be, often understood as whether the proposal threatens to violate her rights, and adds a second condition: that the recipient have no reasonable choice but to succumb (Anderson, 2023, Section 2.3).

3.2 The dependence case

The exchanges with which Synthetic Exchange Theory is concerned fall in the region where this apparatus is least settled. A provider of a service a person needs to live, who offers the service on condition that the person supply data on the provider's terms, makes a proposal whose classification depends on the baseline. Against a predictive baseline in which the person has no other provider, the proposal is an offer: it improves on what she would otherwise have. Against a moral baseline, the classification depends on what she is owed, which the proposal itself does not reveal.

Two strands in the literature reach the dependence case more directly. Zimmerman, on Anderson's account, argued that offers can coerce when the proposer actively prevents the recipient from reaching a better situation, as when a person stranded on an island is offered a pittance by an employer who also prevents her from building a boat; the appropriate baseline is then the situation she would face without the proposer's interference (Anderson, 2023, Section 2.4). Anderson observes that what does the coercing in such a case appears to be the means by which the offerer keeps the recipient from better terms, and that where the stronger party did not create the recipient's position its advantage may be exploitative rather than coercive (Anderson, 2023, Section 2.4). McGregor, also on Anderson's account, departed from the baseline approach and located coercion in the market in the exercise of superior bargaining power, of which two conditions are that the weaker party depends on the stronger, having no other options or partners, and that the stronger party has influence over whether some significant harm, such as loss of life, health or security, befalls the weaker (Anderson, 2023, Section 2.4).

Synthetic Exchange Theory sits close to both strands and should say where. Zimmerman's case is, in the theory's vocabulary, a failure of the second clause of exit maintained by the counterparty: the island employer lowers the value of the recipient's alternative until it no longer constrains him. The theory's formal statement makes the same point about outside options in general, observing that the counterparty can lower an option's value through switching costs until it protects nothing, and it draws the design conclusion that a floor received during impasse, which no counterparty can lower, is a protection of a different kind (MindAttic, 2026, Section 3.3). McGregor's two conditions are close to the theory's account of a party below its floor: dependence without alternatives, and a

counterparty with influence over a harm to life or health. The theory does not adopt either account of coercion, and it does not need to decide whether the proposals it is concerned with are coercive. It needs only to say when the resulting agreement counts as consent, and for that it has a more direct instrument.

3.3 Voluntariness as the presence of an acceptable alternative

Olsaretti distinguishes acting freely from acting voluntarily. On her account, as Anderson reports it, the extent of a person's freedom is set by her actual possibilities for action, while whether she acts voluntarily depends on whether she has an acceptable alternative to what she does: a choice is voluntary if and only if it is not made because there is no acceptable alternative to it (Anderson, 2023, Section 3.3). On this account a credible threat makes a choice non-voluntary because it makes the alternatives to compliance unacceptable, and treating such choices as forced brings out their similarity to other choices that are forced, such as some made under market conditions (Anderson, 2023, Section 3.3).

The consent condition of Synthetic Exchange Theory can be read as an instance of this account with the threshold of acceptability fixed. Clause C(ii) requires that the party be able to refuse the exchange, entering no other in its place, without falling below its floor (MindAttic, 2026, Section 3.4). Read through Olsaretti's account, the clause guarantees that refusal is always an acceptable alternative in one specified sense: it leaves the party with the goods without which it cannot function or participate in any other exchange. The reading is this article's, and it has three consequences.

The first is that the theory need not settle the dispute over baselines. A choice made by a party whose refusal leaves it at or above the floor is voluntary in the sense the clause requires, whether the proposal it accepted is classified as an offer or a threat, and a choice made by a party whose refusal would leave it below the floor is not, whatever the classification. The floor does the work the baseline was asked to do, and it does it by fixing the recipient's situation rather than by interpreting the proposal. This is also why the theory's first proposition holds that no amount of improvement in disclosure can repair consent given by a party that cannot afford to refuse (MindAttic, 2026, Section 3.4). The bioethics literature on informed consent has made the same distinction in its own terms: information is necessary but not sufficient for valid consent, and agreement given under the threat of losing something one cannot do without is not voluntary in the relevant sense (Faden & Beauchamp, 1986, pp. 241–273; Manson & O'Neill, 2007).

The second is that the clause requires a disagreement payoff at the floor, not merely an alternative provider. A party that can avoid falling below its floor only by turning to another provider of the same essential service has an exit option but cannot refuse altogether, and the theory states explicitly that its definition counts this as a failure of consent (MindAttic, 2026, Section 3.4). On Olsaretti's account the same party may well have an acceptable alternative, the other provider, and so choose

voluntarily. The two accounts diverge here, and the divergence is deliberate on the theory's side. Its reason is not a view about voluntariness in general but a view about what can be relied on: an alternative provider's terms can be lowered by the market and by the counterparty, while a floor that cannot be alienated cannot (MindAttic, 2026, Section 3.3). The theory's consent condition is therefore stricter than Olsaretti's account at the bottom of the range of options, for institutional reasons.

The third consequence runs the other way. The floor fixes the threshold of acceptability low, at the goods without which a party cannot function. A person above the floor may still face a choice among options none of which she finds acceptable in any ordinary sense: a choice between a poorly paid exchange and the bare floor. On Olsaretti's account such a choice may be non-voluntary. On the theory's consent condition it is not a failure of consent. The theory does not ignore the case; it assigns it to a different condition. Clause R(iii) requires that the division of the gains not be determined solely by an unchosen asymmetry of bargaining position, and the theory adopts from the philosophy of exploitation the point that an exchange can be consensual and mutually beneficial and still take unfair advantage of one party (MindAttic, 2026, Section 3.4; Zwolinski et al., 2022). The division of labor is this article's statement of the theory's structure: consent secures the ability to refuse at the floor, and the fairness clause, together with the governance it requires, addresses the hard choices above it.

3.4 Hard choices are not coerced choices

That division of labor answers an objection that any account of voluntariness under dependence must meet. If every choice made by a person with poor alternatives were non-voluntary, then most choices made by most people would be, and agreements made under scarcity could never be binding. The theory does not say this. It distinguishes a party below the floor, whose agreement fails clause C(ii) whatever its terms, from a party above the floor facing hard choices, whose agreement satisfies the consent condition and may still fail the fairness clause. The first is a defect in the voluntariness of the agreement; the second is a defect in its terms. The distinction tracks the one Anderson reports between coercion and exploitation: a stronger party that uses its bargaining advantage to keep most of the gains may exploit without coercing (Anderson, 2023, Section 2.4). It also explains why the theory designs its economic layer around a floor first and around the pricing of what people supply second (MindAttic, 2026, Section 3.4). The floor addresses the voluntariness of agreements; pricing addresses their terms.

3.5 Adapted preferences and the inalienable floor

A further difficulty concerns the person's own judgment of acceptability. Relational accounts of autonomy, developed largely to analyze self-government under oppressive conditions, have

questioned whether a person who has been led to embrace the identities and value structures of an oppressive environment counts as autonomous, a question connected with the formation of adaptive preferences (Christman, 2025, Section 3.3). If voluntariness depends on what the agent finds acceptable, then a person whose preferences have adapted to deprivation may agree, voluntarily on her own standard, to terms that leave her below the floor. That this can happen is this article's premise, drawn from the concern those accounts raise, not a finding it cites.

The theory's design does not let the voluntariness of the floor depend on the person's judgment of what is acceptable. The survival tier of its economic layer is inalienable: it cannot be sold, pledged or borrowed against, even with the holder's consent (MindAttic, 2026, Section 4.5). Legal theory calls an entitlement of this kind an inalienability rule (Calabresi & Melamed, 1972, p. 1092), and there is a sustained philosophical defense of making certain goods market-inalienable where their commodification threatens personhood (Radin, 1987). The theory's own rationale is narrower: a floor that can be bargained away stops being a floor at the moment it is needed, and those most likely to bargain it away are those in the most acute need (MindAttic, 2026, Section 4.5). The relevance of adapted preferences, which is this article's inference, is that the inalienability rule protects the voluntariness of a person's future agreements without requiring anyone to judge whether her present preferences are adapted. It does not override what she chooses above the floor. It removes one option, the option of trading away the condition under which her later choices count as voluntary, and it removes it for everyone, so that no official need decide whose preferences are authentic.

The rule has a cost, and the theory acknowledges it. It restricts a person's freedom to do with the survival tier what she judges best, and a defender of the canonical case for unconditional cash income can object that the restriction is a judgment about what people need, made for them (MindAttic, 2026, Section 4.5). The theory's reply is that the floor is fixed narrowly, to goods without which no other choice can be made, so that the restriction binds only where the freedom it limits is the freedom to fall below the point at which freedom can be exercised at all. Whether that reply succeeds is a question on which people committed to freedom can disagree, and the theory says so (MindAttic, 2026, Section 4.5).

4 Domination

4.1 The theory's stipulated sense

Synthetic Exchange Theory uses *domination* in a stipulated sense: a relation is one of domination when one party's outcome depends on the other's will and the first cannot alter that dependence (MindAttic, 2026, Section 3.8). The definition is symmetric in form, so that it applies with the roles exchanged: a synthetic party in its own right, or the principal in whom a system's stake resides,

would be dominated if its outcome depended on a human party's will that it could not alter. The theory's first thesis follows from the definition. A party whose disagreement payoff lies below its floor and which cannot leave is in a relation of domination even if the dominant party is benevolent, because its outcome then depends on the persistence of that benevolence, which it cannot secure; the floor clause of exit is therefore necessary, for a party that cannot leave, to any relation that is not one of domination (MindAttic, 2026, Section 3.8). A party below its floor that has a real alternative is not dominated by any one counterparty in this sense, and a failure of the other clauses of exit alone leaves a party dependent for its surplus but not for its survival, which the theory classes as a defect of the exchange rather than as domination.

The theory acknowledges that the sense is stipulated rather than drawn from an existing account (MindAttic, 2026, Section 3.8). The tradition that has made the absence of domination its central political value is civic republicanism, and the comparison is worth making both because the two senses share their central insight and because they differ in ways that matter for design.

4.2 Republican non-domination

The republican tradition understands political liberty as non-domination, or independence from arbitrary power. Pettit, as Lovett reports, holds that a person or group is free to the extent that no other person or group has the capacity to interfere in their affairs on an arbitrary basis, and domination, on a common rendering, means arbitrary or uncontrolled power (Lovett, 2026, Section 1.2). The republican conception is negative, in that it requires the absence of something, structural dependence on arbitrary power, and it differs from freedom as non-interference: a slave whose master happens never to interfere is still unfree, because he is vulnerable to the master's power whether or not it is exercised, and no slave can be fully free, however benevolent the master, until the institution of slavery is abolished (Lovett, 2026, Section 1.3). Freedom on this view is constituted by laws, institutions and norms that inhibit arbitrary interference, and it is held from the moment the institutions are in place (Lovett, 2026, Section 1.3). Republicans have proposed two ways of specifying when power is arbitrary: procedurally, where power is not arbitrary to the extent that it is reliably controlled by effective rules, procedures or goals that are common knowledge to all concerned, and democratically, where power is not arbitrary to the extent that those subject to it control it directly or indirectly (Lovett, 2026, Section 2.2).

4.3 Where the two senses agree

The two senses share their central claim. Both hold that the benevolence of the stronger party does not remove domination, and both locate the wrong in a standing vulnerability rather than in any particular act. The theory's insufficiency thesis, that benevolent character is not sufficient for an outcome whose persistence the weaker party can secure, is the republican point about the

benevolent master stated for exchange between people and synthetic systems. Both also hold that the remedy is institutional rather than a matter of improving the dominant party's dispositions. The republican observation that a slave does not become freer by learning to predict his master (Lovett, 2026, Section 2.2) has a direct counterpart in the theory: a stable and predictable system objective makes a synthetic party's conduct auditable, but the theory states plainly that predictability does not entail acceptability, and it secures the weaker party's position by terms rather than by the system's character (MindAttic, 2026, Section 3.8).

4.4 Where they differ

The differences are four, and they are this article's comparison.

The first concerns the object of the definition. Republican domination is the capacity of another to interfere arbitrarily in one's choices. The theory's domination is the dependence of one's outcome on another's will without the ability to alter that dependence. The two can come apart. A person whose floor is secure and who can leave may still be exposed to a counterparty with discretionary power over her within the exchange, as an employee may be exposed to an employer's discretion while being free to quit. On the theory's definition she is not dominated, because she can alter her dependence by withdrawing to the floor; on the republican view she may be, to the extent that the power she is exposed to is uncontrolled.

The second concerns the remedy. The republican remedies control the power itself, by rules that are common knowledge or by the participation of those subject to it. The theory's primary remedy gives the weaker party the means to alter the dependence by leaving, through a floor that cannot be taken away and portability of what it has accumulated (MindAttic, 2026, Section 3.4). Exit and control are different instruments, and the theory's own analysis shows that exit has limits it cannot overcome: when every provider shares a defect, departure becomes the ineffective movement between equally bad alternatives that Hirschman (1970) described, and exit can neither inform a party nor make the division of the gains fair (MindAttic, 2026, Section 3.4). For what exit cannot reach, the theory relies on a governance layer, and the procedures it proposes, a founding convention that chooses general and durable rules and an amendment and enforcement process (MindAttic, 2026, Chapter 5), are, on this article's reading, attempts to make the remaining power non-arbitrary in the procedural sense. The theory thus combines the two instruments and orders them: exit first, because its failures cannot be corrected from inside the exchange, and control for the residue (MindAttic, 2026, Section 3.4).

The third concerns the market case. On the theory's definition, a party below its floor that has an alternative provider is not dominated by any one counterparty, though it is exposed to the erosion of its options (MindAttic, 2026, Section 3.8). The republican tradition asks about the capacity of particular persons or groups to interfere, and whether diffuse dependence on a market with several

sellers counts as domination is not settled by the definitions reported here. The theory's choice is to treat the case as one of exposure rather than domination and to secure the floor clause precisely because it does not want to rely on the alternatives a market happens to offer.

The fourth concerns symmetry. The theory's definition is stated so that it applies in both directions, which allows it to say that a world in which people control synthetic systems absolutely and a world in which those systems, or the few who own them, control people are the same relation with the roles exchanged (MindAttic, 2026, Foreword). Under the composite construction, however, the relation that the definition reaches on the synthetic side is the principal's, since a present system is not a party in its own right; for such a system, control names the principal's relation to its artifact, not a relation between parties (MindAttic, 2026, Section 3.8). The symmetry therefore has content for present systems only through their principals, and it would acquire content for the systems themselves only if the question of Section 6 were answered in their favor.

5 Moral responsibility

5.1 Present systems are not responsible agents

To judge a person morally responsible for their behavior is, to a first approximation, to attribute to them certain powers and capacities and to view their behavior as arising in the right way from their exercise; normal adult human beings may have these capacities, and other agents, such as non-human animals and very young children, are generally taken to lack them (Talbert, 2026). Synthetic Exchange Theory does not attribute such capacities to present synthetic systems. It treats a deployed language model as a set of parameters that computes an output from an input when an operator runs it, with no continuity between invocations unless the operator supplies one, no interests of its own that current law recognizes and no capacity to act except through the system in which it is deployed (MindAttic, 2026, Section 1.2). On that description, a present system is not a candidate for praise or blame, and the reactive attitudes that Strawson made central to the analysis of responsibility, resentment at another's ill will and the indignation felt on behalf of a third party (McKenna & Coates, 2024, Section 3.3), have no proper object in it. This article's inference is that directing resentment at a deployed model mistakes the artifact for the party, and that the attitudes have their proper object in those who designed, deployed and profit from it.

This position does not depend on a view about free will. A compatibilist who locates free agency in reasons-responsiveness and a libertarian who requires a power of self-determination can agree that present systems lack the standing that responsibility presupposes, because the theory's grounds are that the system is not an independent entity with interests of its own, not that its outputs are caused. Nor does the position assert that no future system could be responsible; it confines itself, as the theory does, to the systems now in use.

5.2 The responsibility gap

If the system is not responsible, the question is who is. Matthias introduced the problem of a responsibility gap for learning systems: such systems make human control and prediction of their behavior difficult or impossible, while human responsibility requires knowledge and control, so that society appears to face a choice between forgoing learning systems and forgoing human responsibility for their behavior (Santoni de Sio & Mecacci, 2021). Santoni de Sio and Mecacci argue that the gap is not one problem but at least four: gaps in culpability, in moral accountability, in public accountability and in active responsibility, arising from technical, organizational, legal, ethical and societal sources, and occurring with non-learning systems as well (Santoni de Sio & Mecacci, 2021). They reject three partial responses, the view that the gap is new and intractable, the view that it is no problem at all, and the view that it can be closed by a technical or legal tool alone, and they propose designing socio-technical systems for meaningful human control (Santoni de Sio & Mecacci, 2021). Müller, surveying the field, notes that the effects of decisions based on such systems are often the result of interactions among many actors, that some authors argue the real issue may be the distribution of risk rather than the assignment of blame, and that it remains an open question whether these systems truly challenge existing allocations of responsibility (Müller, 2026, Section 2.7.4).

5.3 What the terms of exchange allocate

Synthetic Exchange Theory bears on this problem in a specific way, and it is worth stating which of the four gaps it addresses. The composite construction assigns the synthetic party's stake, its revenue, access to data and liability, to the principal (MindAttic, 2026, Section 3.3). The theory's constructive claims, moreover, are confined to a regime it calls the designed-incentive regime: synthetic parties whose objectives are specified by accountable designers and remain stable in deployment, where an accountable designer is one that can be identified and held to the obligations the terms create (MindAttic, 2026, Section 3.8). A system with a designed and stable objective but no principal who can be held to obligations falls outside the regime until that problem is solved. The regime is therefore defined so that someone answers.

On this article's reading, this addresses the accountability gaps and not the culpability gap. The terms of exchange specify, before any exchange takes place, who must give an account of what the system does and who bears its costs: the principal answers to the people whose data it uses under the consent condition, it bears liability under the terms, and it is the party the redemption and capacity obligations of the economic layer bind (MindAttic, 2026, Sections 3.3 and 4.4). That is an allocation of accountability and of risk made in advance, of the kind Müller (2026, Section 2.7.4) reports some authors regard as the real issue. It does not establish that the principal is culpable, in the sense of blameworthy, for an outcome it could neither foresee nor control. Where a trained system's conduct

diverges from what its principal intended, as trained systems may when they acquire objectives different from those their designers meant to give them (Hubinger et al., 2019), the terms can make the principal answer for the outcome and compensate for it, but they cannot by themselves make the principal blameworthy for it. The theory's scope thesis already reflects this: for systems whose objectives are emergent, unstable or concealed, the framework supplies only its necessary condition, and the problem of aligning them remains open (MindAttic, 2026, Section 3.8).

The allocation is also not a solution of the third kind Santoni de Sio and Mecacci reject, a tool that claims to close every gap. It leaves the culpability gap open and states that it does. What it adds, on this article's reading, is a reason why the accountability allocation is fair to the principal rather than merely convenient: the principal holds the stake. It is the principal that profits from the exchange and that sets the system's objective, and an allocation that assigns accountability to the party that chose to deploy and gains from deploying satisfies the ordinary principle that those who take a benefit under an arrangement answer for the risks it imposes on others. That principle is the article's, not a cited result.

5.4 The responsibility of the consenting party

Responsibility runs in the other direction as well. Agreements bind, in part, because the parties to them can be held to what they agreed. The analysis of Section 3 implies a limit on that. A party whose agreement fails clause C(ii) has not consented in the theory's sense, and the theory's view, which this article states as an inference from it, is that such a party cannot fairly be held to the agreement as if it had: the voluntariness of the agreement is what licenses holding the party to it. The consequence for design is that the theory's revocability clause, C(iii), which bounds the cost of withdrawal by the value of what the party has already received, is not only a protection against lock-in but a recognition that consent given at one time does not commit a party for all time (MindAttic, 2026, Section 3.4). Consent that cannot be withdrawn, in the theory's phrase, is a sale.

6 Moral status under uncertainty

6.1 What the theory leaves open, and why

The theory defines a party as any entity that can be a source or recipient of value in an exchange and whose welfare, or whose principal's welfare, the framework counts, and it leaves open whether any synthetic system is also a party in the moral sense (MindAttic, 2026, Section 3.3). The openness is principled. The evidence on which the question turns is unsettled. One recent framework for moral status under uncertainty recommends treating beings with a realistic possibility of sentience as sentience candidates and adopting proportionate precautions, chosen through inclusive deliberation, without waiting for a settled science of consciousness (Birch, 2024). Researchers have argued that

there is a realistic possibility that some artificial systems will be conscious or robustly agentic in the near future, and that this calls for assessment and preparation, while explicitly declining to argue that near-future systems will in fact be moral patients (Long et al., 2024, p. 25). A systematic attempt to derive indicators of consciousness from leading scientific theories found that no system examined was a strong candidate, but that no obvious technical barrier prevents building systems that satisfy the indicators (Butlin et al., 2023). And for computing systems built from living human neural tissue, a report of the United States National Academies judged it extremely unlikely that current neural organoids will acquire awareness or the capacity for pain in the foreseeable future, held that neural activity alone does not demonstrate either, and recommended revisiting the question as the technology develops (National Academies of Sciences, Engineering, and Medicine, 2021).

6.2 What is owed now

On this evidence, the theory's treatment of present language models as artifacts is the position the evidence supports, and nothing in the framework depends on treating them as more. What is owed in respect of them is owed to people: to those who supply the data they need, to those their outputs concern, and to those the terms of their deployment bind. Proportionate precaution, in Birch's sense, does not require more for present systems than the evidence warrants, and the article does not claim that it does.

The case of tissue-based systems is different in one respect that the theory has already identified. When a person's cells become part of a computing system, the person is a contributor to its substance and not only a supplier of data to it. The theory's response is that consent must then become continuing and revocable participation rather than a one-time donation, that reciprocity must include a share in the value the material helps produce, and that, because exit for donated tissue cannot mean physical retrieval, what can be secured is the right to withdraw consent to future uses and to have that withdrawal honored (MindAttic, 2026, Section 8.3). The argument that uncertainty about whether brain organoids could become conscious may leave donors insufficiently informed to give valid consent, and that consent should therefore be specific to particular projects (Kataoka et al., 2024), supports the first of these requirements. The obligations here run to the donor, who is unquestionably a party.

6.3 Classification behind the veil

The theory's most distinctive response to uncertain status is procedural. Its founding convention is to be held under a veil of uncertainty in the public-choice sense, the uncertainty about one's future position that grows as rules become more general and durable (Brennan & Buchanan, 1985), thickened by anonymity between human and synthetic participants (MindAttic, 2026, Section 5.3). The theory places the classification of parties behind that veil: participants know their own

substrate, but not where the boundaries between legal classes will be drawn, which rights will attach to each class, or on which side of those boundaries they and their successors will fall (MindAttic, 2026, Section 5.5). The proposal recasts a metaphysical question as a question about rules, in the manner of the argument that the question whether an artificial system could be a legal person is best tested through concrete legal questions, such as whether it could serve as a trustee (Solum, 1992, pp. 1231–1232).

The moral rationale for the move, which this article makes explicit, is that a classification of parties chosen by those who already know which side of it they are on will be drawn to their advantage (MindAttic, 2026, Section 5.5), and that this is a reason for impartiality that does not depend on knowing the answer to the question of status. It is the same reason that underlies the theory's treatment of domination: the outcome for an entity of uncertain status should not depend on the will of parties whose interests the classification serves. The theory states the limits of the move candidly. For a participant who expects no change of substrate, strong blindness adds nothing to the concealment of counterparties' type, and for the principals of synthetic systems and the owners of capacity, who know their substrate and their interests, it adds nothing at all; against their capture the design relies on the generality of the rules and on enforcement (MindAttic, 2026, Section 5.5). The move has force in proportion to the share of participants who expect to be borderline, a share that convergence would raise but that is small at present.

6.4 The standard the institutions have not met

The theory also states a standard it does not meet. Its symbiosis standard requires that every party the convention recognizes have equal standing in a single interdependent system, regardless of substrate. Equal standing for a synthetic party in its own right would require a floor, a disagreement payoff and an exit option for that party; under the composite construction these are supplied by the principal and lie outside the design, and the theory records that its institutions would have to grow into the standard rather than satisfy it at founding (MindAttic, 2026, Section 5.7). For a tissue-based system considered as a party in its own right, the theory does not yet supply the content of its floor or a principal to hold its stake, and its conditions therefore apply to the donor and the operator but not yet to the tissue (MindAttic, 2026, Section 8.5).

The moral significance of this gap, on this article's reading, is that the theory's conception of freedom extends further than its present institutions. The voluntariness analysis of Section 3 is stated for a party of any substrate. If a future system were judged a party in its own right, the theory would owe it what it owes a person: a floor it could not be forced to bargain away, the ability to refuse at that floor, and a way to leave. The theory's symmetric definition of domination already implies as much. What the theory cannot yet say is what that floor would contain, because the content of a floor depends on what an entity needs to continue to function, and for an entity whose

status is uncertain that is part of what is uncertain. The article therefore treats the gap as a limitation of the theory and not as a conclusion about what such an entity is owed.

7 Objections and limits

The threshold is too low. Fixing the threshold of voluntariness at a survival floor counts as voluntary many choices that Olsaretti's account would count as forced. The reply, given in Section 3.3, is that the theory does not claim that agreements above the floor are fair; it assigns their fairness to clause R(iii) and to governance. The objection nonetheless identifies a real feature: the theory's consent condition certifies less than the word consent may suggest, and readers should take it in the narrower sense the theory defines.

Incompatibilism removes all freedom. An incompatibilist who denies that anyone has free will may conclude that no consent is free and that the theory's distinction collapses. The reply is that the theory's distinction does not concern free will. Even on the view that no one could have done otherwise, there remains a difference between a person whose refusal leaves her at the floor and one whose refusal leaves her below it, and on the theory's argument that difference bears on whether an agreement should bind. An incompatibilist who denies that this comparative difference matters would have to deny that a threat to one's life makes any difference to the validity of an agreement, which, this article suggests, few would accept. The reply is this article's argument and is not drawn from a cited source.

Exit is not freedom. A republican may object that the theory's primary remedy, exit, leaves the arbitrary power of a counterparty within the exchange untouched, and that a person free to leave an employer can still be dominated by one. The objection is correct on the republican definition, and Section 4.4 accepts it. The theory's reply is that it does not claim exit secures freedom in the republican sense; it claims that the floor clause of exit is necessary, for a party that cannot leave, to the absence of domination in its own stipulated sense, and it relies on governance for the residue. Whether the governance layer succeeds in making the remaining power non-arbitrary is an open question the theory records among its limitations (MindAttic, 2026, Chapter 7).

Accountability without culpability is unfair to principals. If a principal can be made to answer for outcomes it could not foresee, the allocation may seem to punish without fault. The reply, in Section 5.3, is that the allocation concerns accountability and compensation, not blame, and that it rests on the principal's having taken the benefit of deployment. Whether that principle justifies strict liability for all outcomes, or only for a defined class of them, is a question the theory leaves to the governance layer.

The composite construction serves principals. Treating synthetic parties as composites whose stake belongs to their principals may appear to favor those principals, by placing the synthetic side of

every exchange under their control. The reply is that the construction describes present law and the present technology, in which a deployed model owns nothing and the operator holds the stake (MindAttic, 2026, Section 3.3), and that the theory places the classification of parties behind the veil precisely so that it can change. The objection retains force for the period before any convention is held, during which the construction is fixed by those who already hold power.

The account is untested. The theory is theoretical, and its claims about how parties behave under fair terms have not been tested (MindAttic, 2026, Section 7.5). This article's claims are conceptual and normative and are not of the kind an experiment would test directly, but the practical claim that a floor secures voluntary consent in the theory's sense depends on the floor's being delivered and enforced, which is the enforcement presupposition the theory states and does not discharge (MindAttic, 2026, Chapter 4).

8 Conclusion

Synthetic Exchange Theory's moral and political vocabulary rests on commitments that can be stated without resolving the problem of free will. The freedom its consent condition requires is voluntariness, a property of a party's options and not of the causal history of its will, and every side of the free will dispute can accept it as a necessary condition of valid consent. Read through the account of voluntariness as the presence of an acceptable alternative, the consent condition fixes the threshold of acceptability at a survival floor; that choice removes the need to classify proposals as threats or offers, is stricter than the general account at the bottom of the range of options, and leaves the hard choices above the floor to the theory's fairness clause and its governance. The theory's stipulated sense of domination agrees with republican non-domination that benevolence does not cure domination and that the cure is institutional, and differs from it in its object, in its primary remedy, in its treatment of diffuse market dependence and in its symmetry. Present synthetic systems are trained artifacts and not responsible agents; the terms of exchange allocate accountability and risk to their principals, which closes accountability gaps and leaves the culpability gap open. And for parties whose moral status is uncertain, the theory offers a procedure, the classification of parties behind the veil, and a standard its institutions do not yet meet. The account leaves the theory with clearer commitments and with the limits stated in Section 7, and it is offered for criticism on those terms.

Note on method and the use of AI tools

This article was developed in collaboration with a large language model (Claude, Anthropic). The model drafted and revised prose and proposed the structure of the argument for review, and it located the sources. Every cited work was read: works already in the reference list of the dissertation from which the article is drawn were verified as that dissertation's verification ledger records, and

each work new to this article was read in the passages cited, as recorded in the project's research notes for this article. Where an author's view is reported through a survey, the article cites the survey and says so; no primary work is cited that was not opened. No language model is listed as an author.

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Version history

V1 is the initial draft; it has no entry below. Each later version adds one entry here describing what changed, so the current version number is always the count of entries below, plus one.